

# MLOps

## Environments – Docker – GCP

Session 5P

Juan Ramírez  
jramirezg@faculty.ie.edu

# Environments

# What is a Python Environment?

- Isolation – Python packages and versions live in their own space
- Organization – Each project can maintain it's own requirements file of python packages and you don't have version overlap
- Anti-globalization – Installing stuff to the system globally is generally not a good idea



## Uses:

- Testing (python 2.6, 2.7, 3.3)
- Development
- Trying new packages from PyPI
- Separating deployed apps with different dependency needs
- Trying new versions of Python
- Reproducing someone's work

Easy to make  
Easy to throw away

`conda create -h`

# Manage Environments with Conda

- [Managing environments — conda 4.6.1 documentation](#)

## Create Environment

```
$ conda create -n py3k python=3.6
```

Name of environment

## List Environments

```
$ conda info --envs
```

## List Packages

```
$ conda list (if environment is activated)  
$ conda list -n py3k (if not activated)
```

## Activate Environment

```
$ conda activate py3k
```

## Install Package

```
(py3k) $ conda install scikit-learn
```

## Removing Package

```
(py3k) $ conda remove scipy
```

## Removing Environment

```
(base) $ conda remove -n py3k --all
```

# Pyenv

- <https://github.com/pyenv/pyenv>

pyenv lets you easily switch between multiple versions of Python. It's simple, unobtrusive, and follows the UNIX tradition of single-purpose tools that do one thing well.

```
$ pyenv versions
2.7.10
* 3.5.0 (set by /Users/yyu/.pyenv/version)
  miniconda3-3.16.0
  pypy-2.6.0

$ python --version
Python 3.5.0

$ pyenv global pypy-2.6.0

$ python --version
Python 2.7.9 (295ee98b69288471b0fcf2e0ede82ce5209eb90b, Jun 01 2015, 17:30:13)
[PyPy 2.6.0 with GCC 4.9.2]

$ cd /Volumes/treasuredata/jupyter

$ pyenv version
miniconda3-3.16.0 (set by /Volumes/treasuredata/.python-version)

$ python --version
Python 3.4.3 :: Continuum Analytics, Inc.
```

# Poetry

- <https://python-poetry.org/>

Python packaging and dependency management made easy

```
$ poetry add pendulum

Using version ^2.0.5 for pendulum

Updating dependencies
Resolving dependencies... (1.5s)

Writing lock file

Package operations: 4 installs, 0 updates, 0 removals

- Installing six (1.13.0): Downloading... 25%
- Updating pytzdata (2019.3 -> 2020.4): Installing...
- Installing pendulum (2.0.5)
```

Build and package your projects:

```
$ poetry build

Building poetry (1.0.0)
- Building sdist
- Built poetry-1.0.0.tar.gz

- Building wheel
- Built poetry-1.0.0-py2.py3-none-any.whl
```

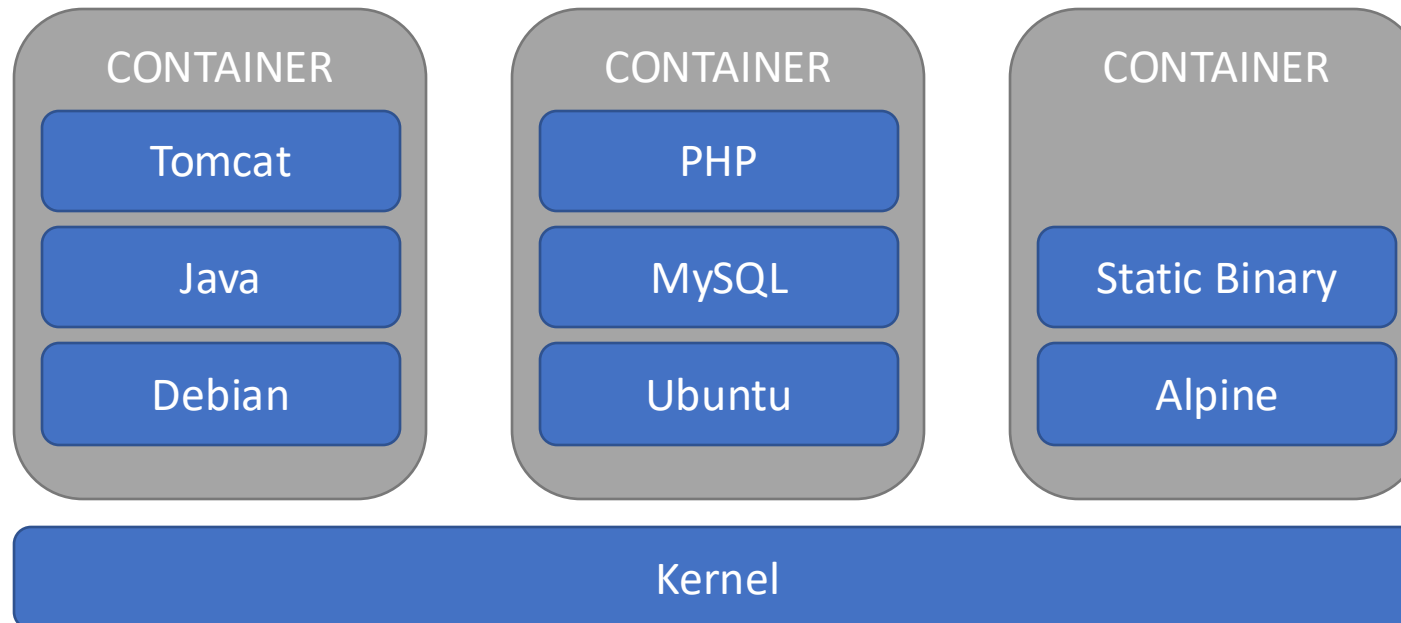
# Docker

# What is Docker?



<https://www.docker.com/>

- Docker is an Open Source platform tool to manage containers
- Containers allow us to build an application with required libraries, binaries and dependencies needed to run the application



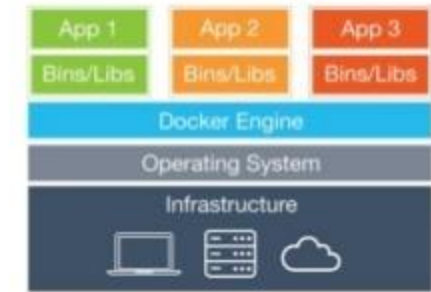
# Why use Docker?

- Portability
- Light weight Operating System
- Standardized packaging for software and dependencies
- Isolate apps from each other
- Share the same OS kernel
- Resource optimization
- Continuous Deployment and Testing

## Containers vs. VMs



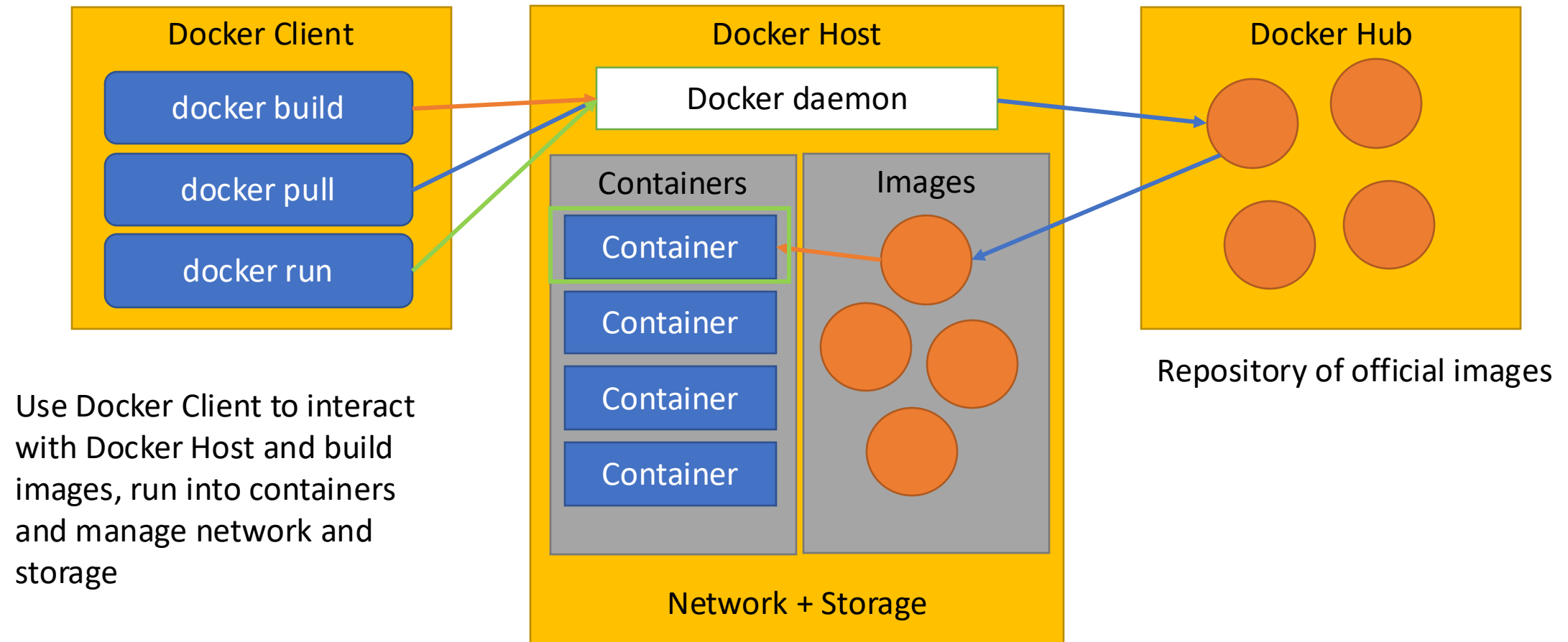
Virtual Machines



Containers

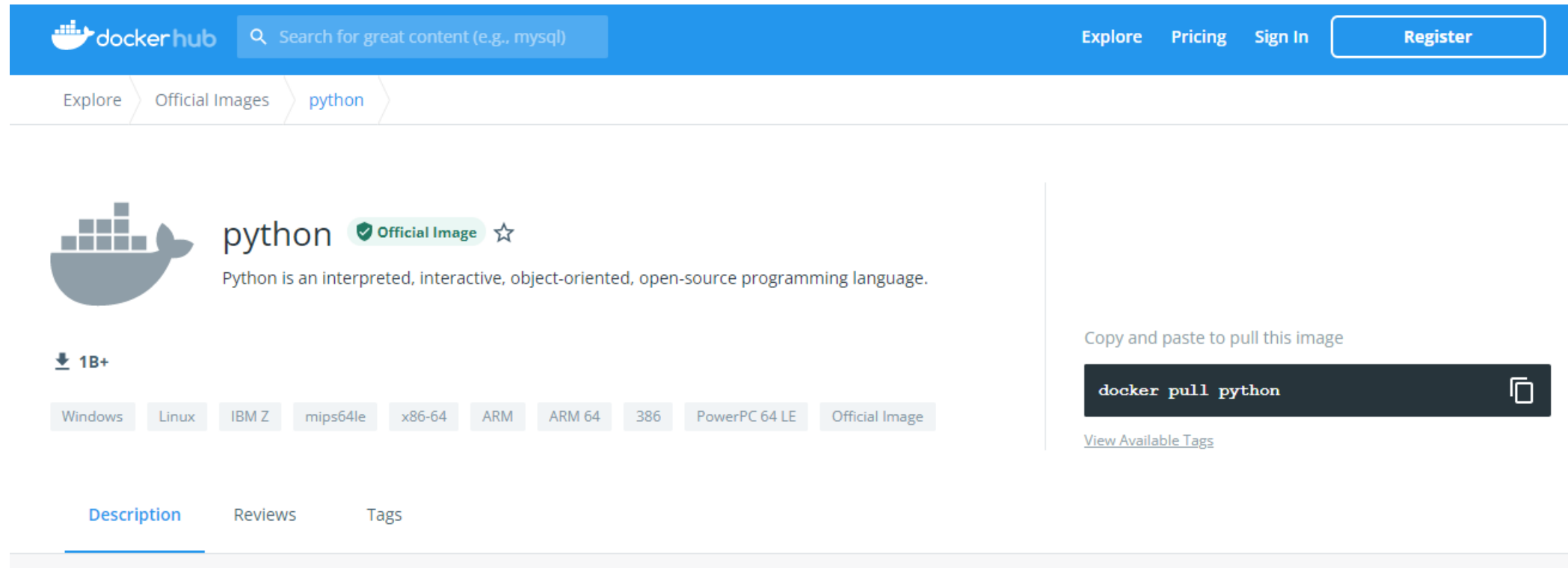


# Docker Architecture



# Docker Hub

- [https://hub.docker.com/\\_/python/](https://hub.docker.com/_/python/)



The screenshot shows the Docker Hub interface for the 'python' image. At the top is a blue navigation bar with the Docker Hub logo, a search bar, and links for Explore, Pricing, Sign In, and Register. Below the navigation bar is a breadcrumb trail: Explore > Official Images > python. The main content area features the 'python' image with a Docker logo icon, a green 'Official Image' badge, and a star icon. The description states: 'Python is an interpreted, interactive, object-oriented, open-source programming language.' Below the description, it shows '1B+' downloads. A row of platform filters includes Windows, Linux, IBM Z, mips64le, x86-64, ARM, ARM 64, 386, PowerPC 64 LE, and Official Image. At the bottom, there are tabs for Description, Reviews, and Tags. On the right side, there is a section titled 'Copy and paste to pull this image' with a dark button containing the command 'docker pull python' and a copy icon. Below the button is a link to 'View Available Tags'.

# Basic Docker Client commands

\$ docker image pull <image name>:<version>

\$ docker image ls

\$ docker container run -d -p 5000:5000 --name <container name> <image name>:<version>

\$ docker container ps

\$ docker container stop <container name> (or <container id>)

\$ docker container rm <container name> (or <container id>)

\$ docker image rmi <image name> (or <image id>)

\$ docker build -t <image name>:<version> .

\$ docker image push <image name>:<version>

\$ docker --help

# Dockerfile

- List of instructions to build the docker images

|                                                                            |                                                                                                            |                                                        |
|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| The latest version of Ubuntu is used as the base image for this Dockerfile | FROM ubuntu:latest                                                                                         |                                                        |
|                                                                            | MAINTAINER <a href="mailto:juan.ramirez@mycompany.com">juan.ramirez@mycompany.com</a>                      | Set the author field for the generated image           |
| Run some commands                                                          | RUN apt-get update && apt-get install -yq curl<br>RUN apt-get clean<br>RUN pip install -r requirements.txt |                                                        |
|                                                                            | CMD ["top"]<br>CMD ["ls", "-l"]                                                                            | Commands to run when the generated image gets launched |
| Set working directory and environment variables                            | WORKDIR /root<br>ENV VAR1 version1                                                                         |                                                        |
|                                                                            | ADD run.sh /root/run.sh<br>RUN chmod +x run.sh                                                             | Add a shell script to the generated image              |
| Set entry point and arguments for the generated image                      | ENTRYPOINT ["/run.sh"]<br>CMD ["arg1"]                                                                     |                                                        |

Source: Understand Dockerfile (Rocky Chen, Medium, 2020)

# Requirements file

- List of libraries that we need to install in order to run the model

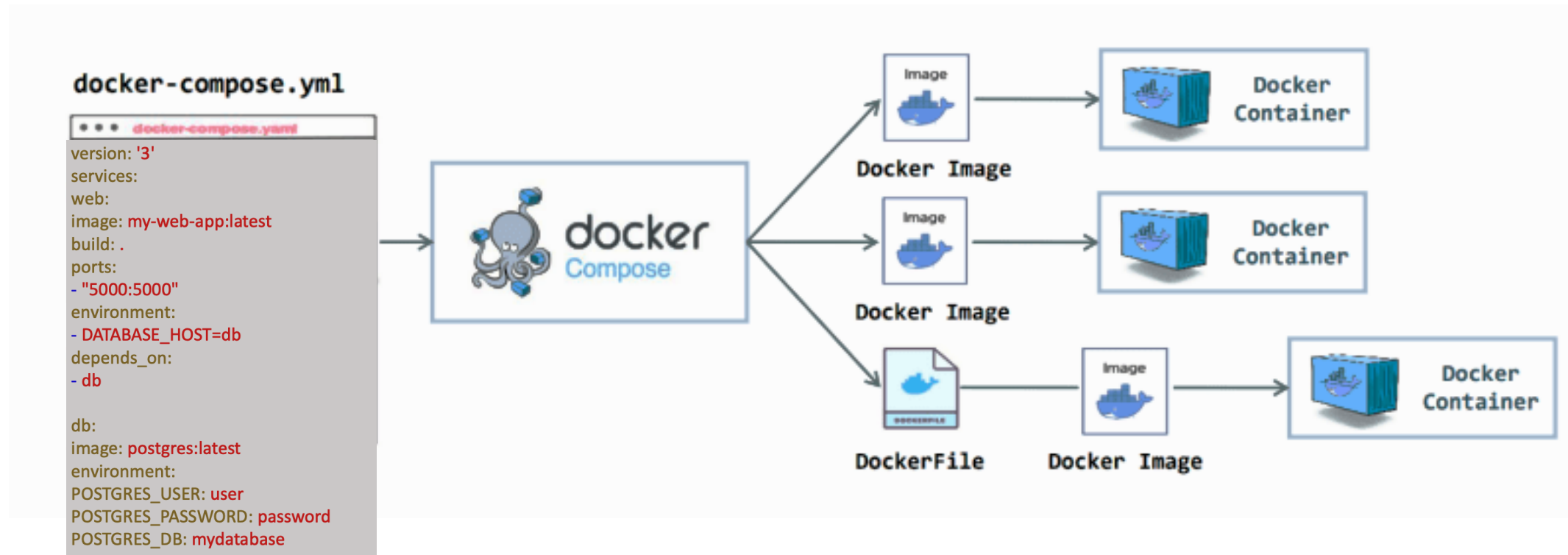
```
requirements.txt
1 pandas
2 scikit-learn
3 flask
4 gunicorn
```

- Just a list or we can include also specific versions scikit-learn==1.0.2

```
mlflow
cloudpickle==2.0.0
dataclasses==0.6
psutil==5.9.0
scikit-learn==1.0.2
typing-extensions==4.1.1
```

# Docker Compose

- Docker Compose is a tool that makes it easy to run multiple containers at once.
- It allows you to define all the containers, networks, and volumes for your application in a single file. This file is called a “docker-compose.yml” file.



Source: What and Why of Docker Compose (Javeriasohail, Medium, 2024)

# Why Docker Compose?

- 1. Simplifies multi-container applications:** Compose makes it easy to manage multiple containers, reducing the complexity of managing individual containers.
- 2. Defines services and dependencies:** Compose allows you to define services, their dependencies, and how they interact, making it easier to manage complex applications.
- 3. Streamlines development and testing:** Compose enables developers to focus on writing code rather than managing infrastructure, making development and testing more efficient.
- 4. Improves collaboration:** Compose files can be shared among team members, ensuring everyone is working with the same configuration.
- 5. Enhances portability:** Compose files are platform-agnostic, making it easy to deploy applications across different environments.
- 6. Simplifies scaling:** Compose makes it easy to scale services up or down as needed.

# docker-compose.yml

Version of the Compose file format

```
version: '3'
```

```
services:
```

```
web:
```

```
image: my-web-app:latest
```

```
build: .
```

```
ports:
```

```
- "5000:5000"
```

```
environment:
```

```
- DATABASE_HOST=db
```

```
depends_on:
```

```
- db
```

Service 1

```
db:
```

```
image: postgres:latest
```

```
environment:
```

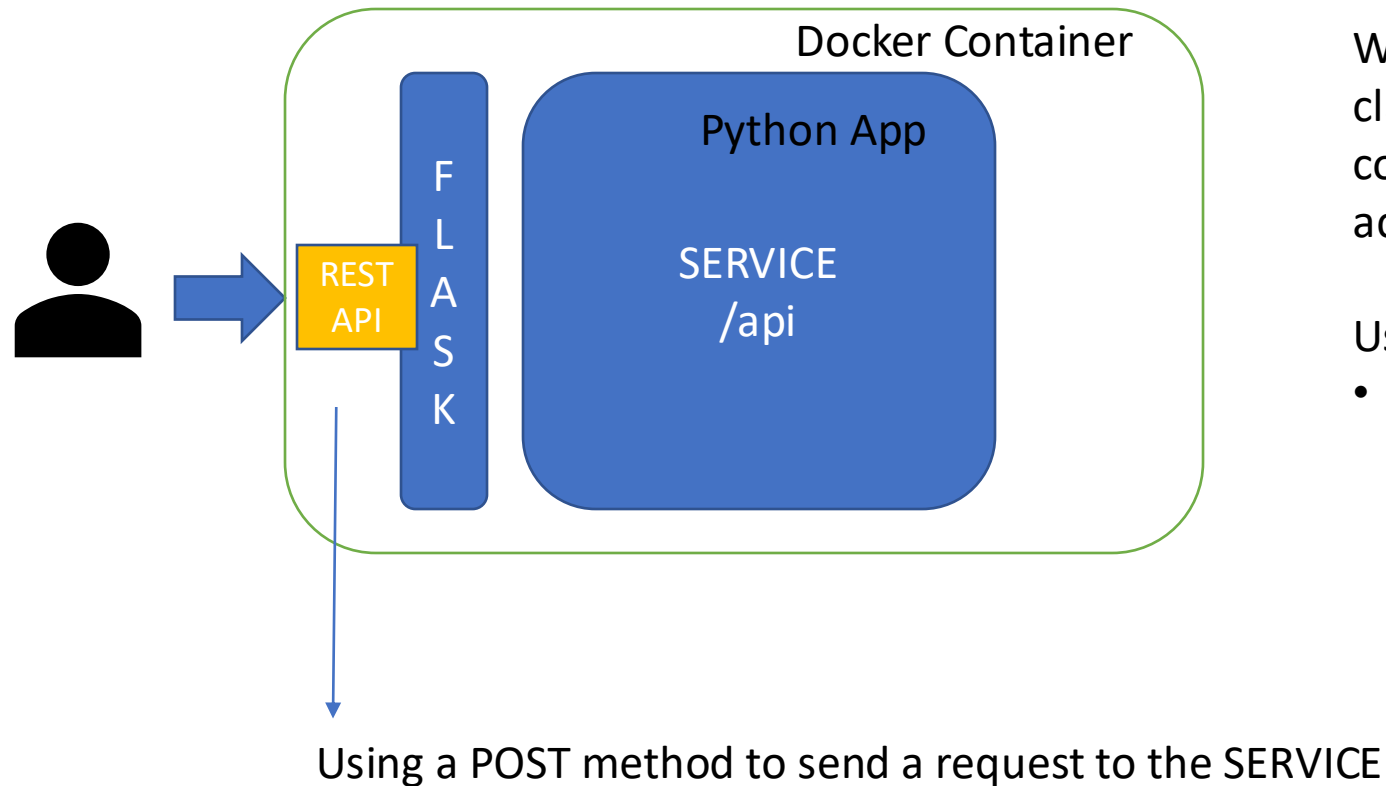
```
POSTGRES_USER: user
```

```
POSTGRES_PASSWORD: password
```

```
POSTGRES_DB: mydatabase
```

Service 2

# Model Deployment in Docker Using Flask



- Use case:

We are going to use IRIS dataset to build a classification model and then deploy it in a docker container and expose it as REST API which will be access over HTTP.

Using Flask python package:

- Flask is used to expose our machine learning model as a REST API. Flask is not a complete production ready server.

# Model Deployment in Docker Using Flask - Steps

1. Train model and save pkl file -> use model.py
2. Create application using flask -> use app.py
3. Create requirements.txt file
4. Create Dockerfile
5. Put model.pkl, app.py, requirements.txt and Dockerfile in the same directory
6. Create docker image -> `docker build -t my_iris:v1 .`
7. Run docker container -> `docker container run -d -p 5000:5000 --name my_iris_app my_iris:v1`
8. Test service
  1. Use requests.py
  2. Use curl
  3. Use talend

<https://github.com/juanmaramirezgarcia/docker-jenkins-flask.git>

# Test Application: Requests Python Library

```
feature = [[4.4, 2.9, 2.4, 4.2]]
```

Vector with values to score

[31] ✓ 0.2s

Python

```
import requests
```

```
url = 'http://127.0.0.1:5000/api'
```

Url

```
r = requests.post(url, json={'feature': feature})
```

Capturing response

```
print(r.text)
```

[34] ✓ 0.2s

Python

... 1

Score

```
labels = {
```

```
    0: "setosa",
```

```
    1: "versicolor",
```

```
    2: "virginica"
```

```
}
```

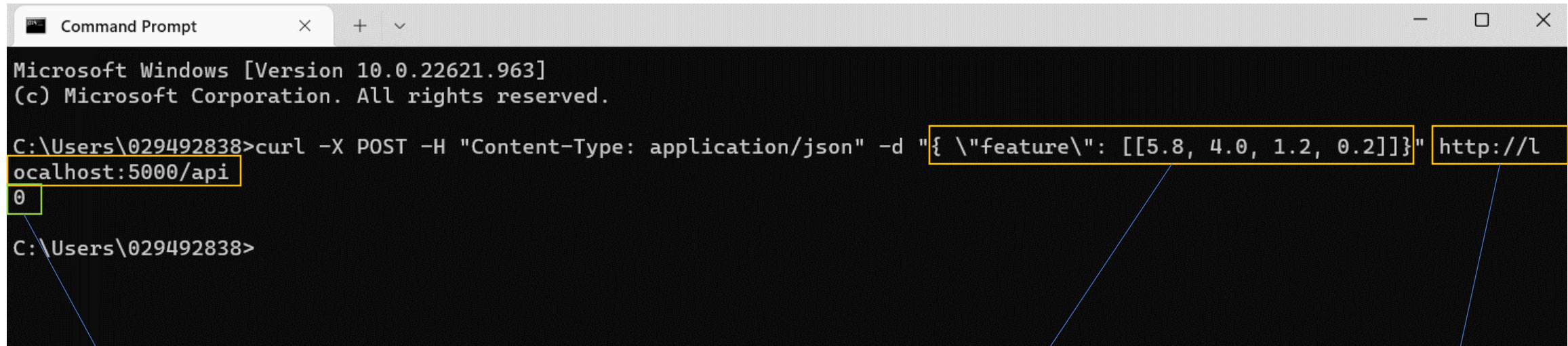
```
print(labels[r.json()])
```

[35] ✓ 0.3s

Python

... versicolor

# Test Application: Curl from command line (Windows)



```
Microsoft Windows [Version 10.0.22621.963]
(c) Microsoft Corporation. All rights reserved.

C:\Users\029492838>curl -X POST -H "Content-Type: application/json" -d "{ \"feature\": [[5.8, 4.0, 1.2, 0.2]]}" http://l
ocalhost:5000/api
0
C:\Users\029492838>
```

Score

Vector with values to score

Url

# Test Application: Talend API Tester

MLOps + Add an environment

## IRIS Docker

METHOD: POST

SCHEME :// HOST [ "\*" PORT ] [ PATH [ "?" QUERY ] ]

Url: `http://localhost:5000/api` length: 25 byte(s) Save Send

QUERY PARAMETERS

HEADERS <sup>1</sup> <sub>1</sub>2

☒ Content-Type : application/json + Add header ? Add authorization

Form ▼

BODY <sup>2</sup>

Text ▼

```
1 {
2   "feature": [
3     [
4       5.8,
5       4.0,
6       1.2,
7       0.2]]
8 }
```

Text | JSON | XML | HTML | Format body | ☒ Enable body evaluation length: 51 bytes

## Response

Cache Detected - Elapsed Time: 35ms

200 OK

HEADERS <sup>1</sup>

pretty ▼

Server: Werkzeug/2.2.2 Python/3.8.16  
Date: Tue, 10 Jan 2023 16:55:33 GMT

BODY <sup>2</sup>

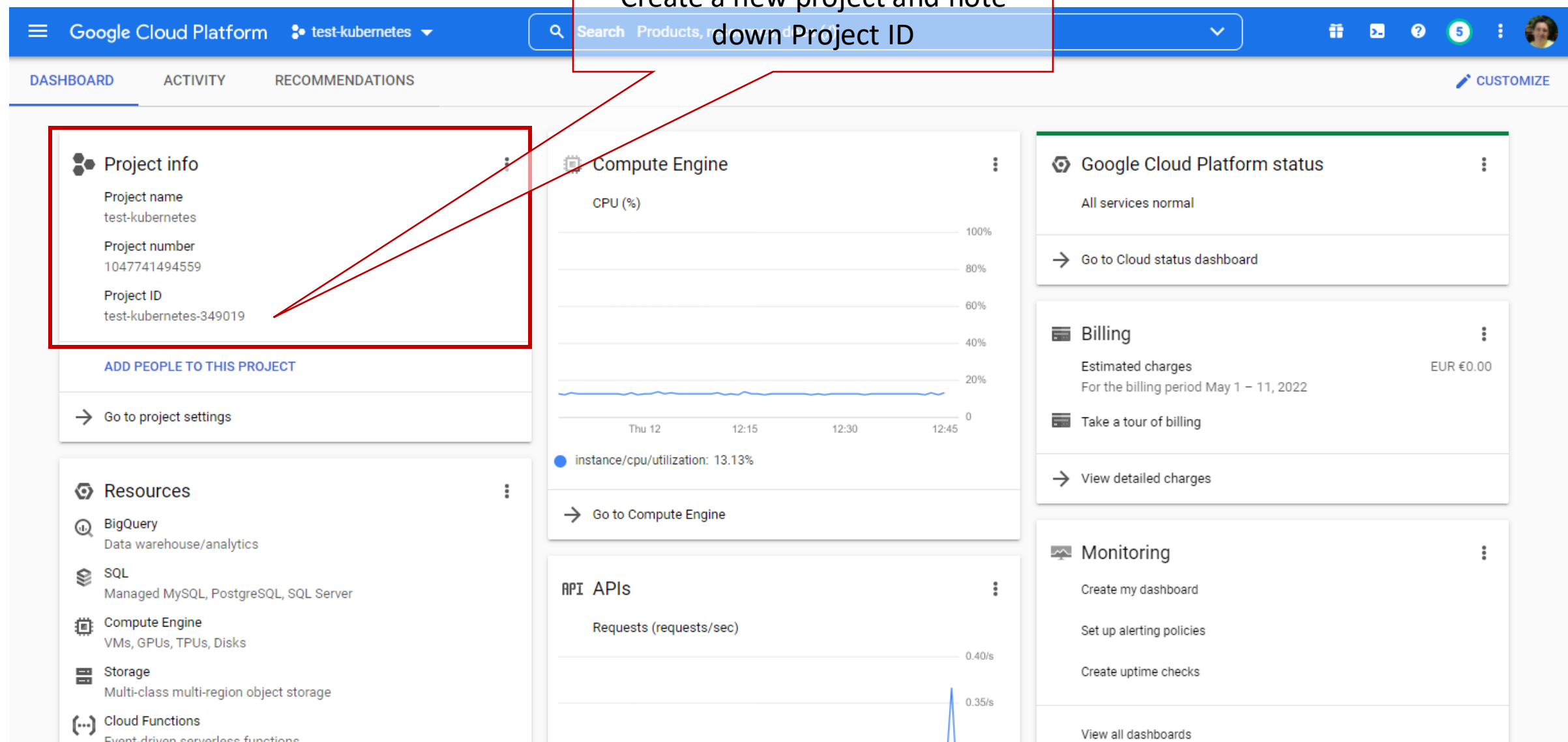
pretty ▼

0

Score

# Deploying Docker in GCP

# GCP Project



**Create a new project and note down Project ID**

**Project info**

- Project name: test-kubernetes
- Project number: 1047741494559
- Project ID: test-kubernetes-349019

[ADD PEOPLE TO THIS PROJECT](#)

[Go to project settings](#)

**Resources**

- BigQuery**  
Data warehouse/analytics
- SQL**  
Managed MySQL, PostgreSQL, SQL Server
- Compute Engine**  
VMs, GPUs, TPUs, Disks
- Storage**  
Multi-class multi-region object storage
- Cloud Functions**  
Event-driven serverless functions

**Compute Engine**

CPU (%)

100%  
80%  
60%  
40%  
20%  
0

Thu 12 12:15 12:30 12:45

● instance/cpu/utilization: 13.13%

[Go to Compute Engine](#)

**APIs**

Requests (requests/sec)

0.40/s  
0.35/s

**Google Cloud Platform status**

All services normal

[Go to Cloud status dashboard](#)

**Billing**

Estimated charges  
For the billing period May 1 – 11, 2022  
EUR €0.00

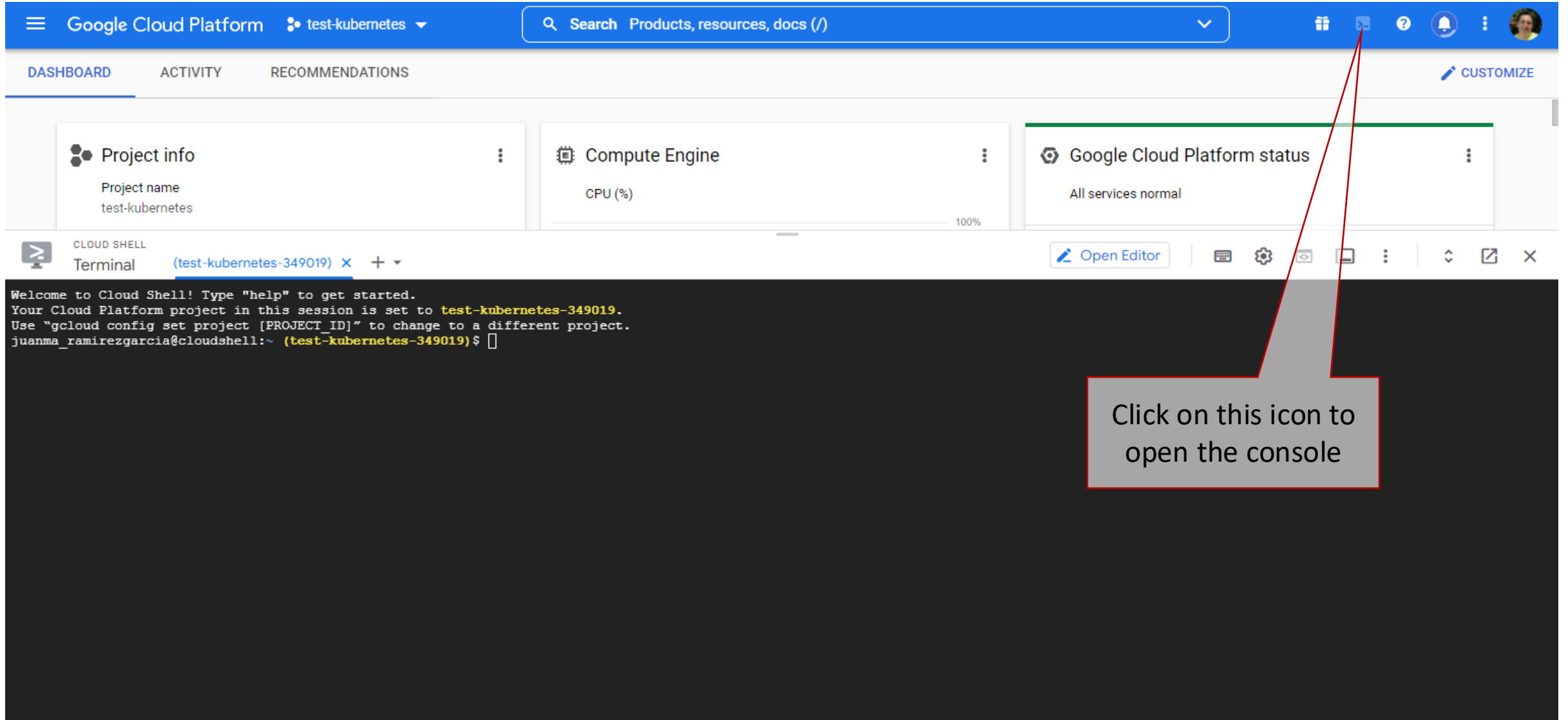
[Take a tour of billing](#)

[View detailed charges](#)

**Monitoring**

- Create my dashboard
- Set up alerting policies
- Create uptime checks
- [View all dashboards](#)

# GCP Project Console



Google Cloud Platform test-kubernetes

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DASHBOARD ACTIVITY RECOMMENDATIONS

CUSTOMIZE

Project info  
Project name  
test-kubernetes

Compute Engine  
CPU (%)  
100%

Google Cloud Platform status  
All services normal

CLOUD SHELL  
Terminal (test-kubernetes-349019) x +

Open Editor

Welcome to Cloud Shell! Type "help" to get started.  
Your Cloud Platform project in this session is set to **test-kubernetes-349019**.  
Use "gcloud config set project [PROJECT\_ID]" to change to a different project.  
juanma\_ramirezgarcia@cloudshell:~ (**test-kubernetes-349019**) \$

Click on this icon to open the console

# Google Cloud Platform Console Commands

- Clone your repository
  - Export your PROJECT\_ID
  - Build image
  - Authorize Configure Docker
  - Push your image to GCR
  - Set PROJECT\_ID
  - Set Compute/Zone
  - Enable container service
  - Create Cluster
  - Create Deployment
  - Expose Deployment
- `git clone https://github.com/juanmaramirezgarcia/house-flask-docker.git`
  - `export PROJECT_ID=test-kubernetes-349019`
  - `docker build -t gcr.io/$PROJECT_ID/madrid:v1 .`
  - `gcloud auth configure-docker`
  - `docker push gcr.io/$PROJECT_ID/madrid:v1`
  - `gcloud config set project $PROJECT_ID`
  - `gcloud config set compute/zone europe-west1-b`
  - `gcloud services enable container.googleapis.com`
  - `gcloud container clusters create madrid-cluster --num-nodes=2`
  - `kubectl create deployment madrid-app --image=gcr.io/$PROJECT_ID/madrid:v1`
  - `kubectl expose deployment madrid-app --type=LoadBalancer --port 5000 --target-port 5000`

Note this are the exact commands in my project. You need to replace with the appropriate values in your project

# Google Cloud Container Registry

- What is Google Cloud Container Registry? <https://cloud.google.com/container-registry>
- Google Cloud Container Registry is a place to manage Docker images, perform vulnerability analysis, and decide who can access what with fine-grained access control.
- Existing CI/CD integrations let users set up fully automated Docker pipelines.



## Secure, private Docker registry

Get access to secure private Docker image storage on Google Cloud Platform in minutes. Maintain control over who can access, view, or download images. Get consistent uptime on an infrastructure protected by Google's security.



## Build and deploy automatically

Automatically build and push images to private registry when you commit code to Cloud Source Repositories, GitHub, or Bitbucket. Easily set up CI/CD pipelines with integration to Cloud Build or deploy directly to Google Kubernetes Engine, App Engine, Cloud Functions, or Firebase.



## In-depth vulnerability scanning

Detect vulnerabilities in early stages of the software deployment cycle. Make certain your container images are safe to deploy. Constantly refreshed database helps ensure your vulnerability scans are up-to-date with new malware.



## Lock down risky images

Use native integration with Binary Authorization to define policies and prevent deployment of images conflicting with the set policies. Trigger automatic lock-down of container images to prohibit risky images from being deployed to Google Kubernetes Engine.



## Native Docker support

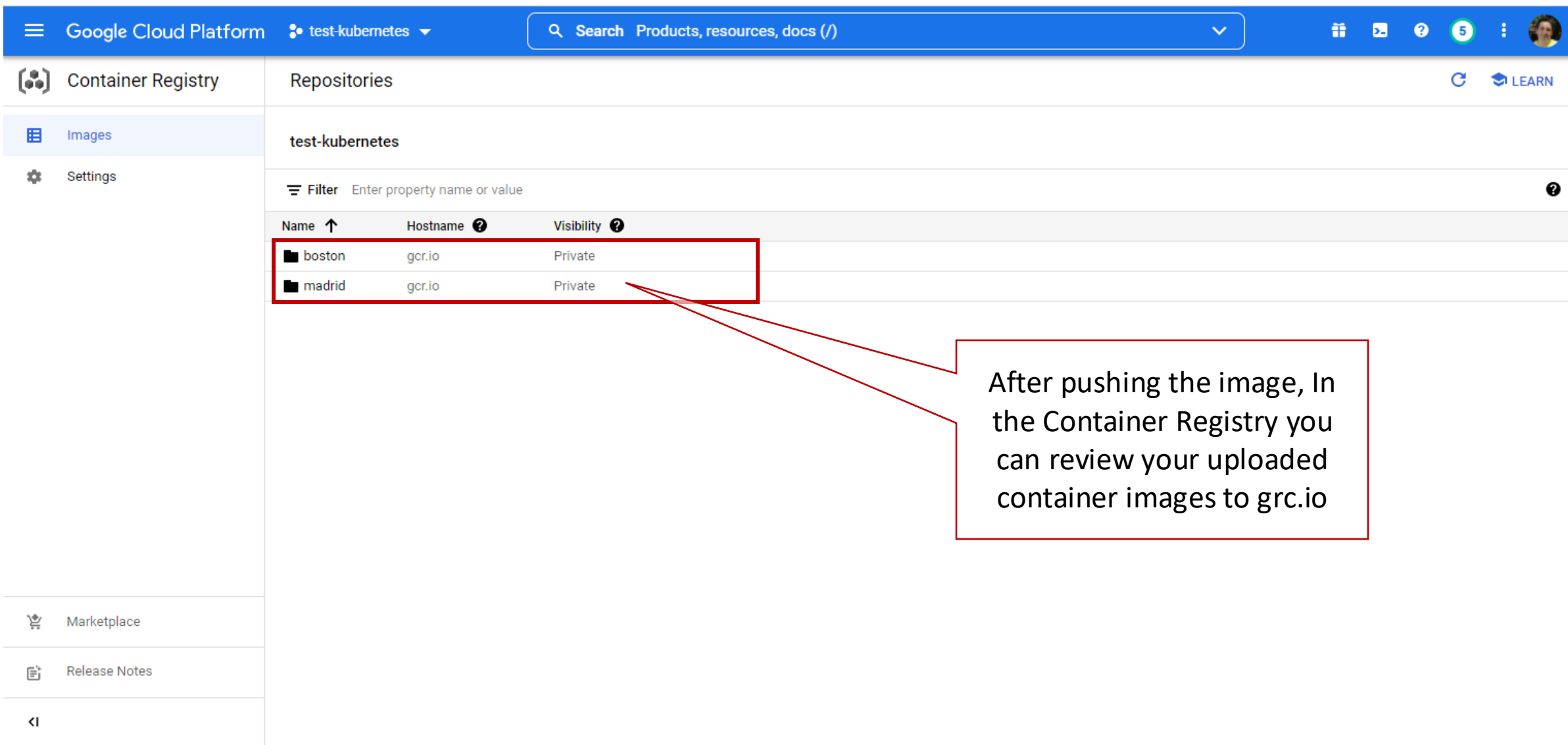
Define multiple registries based on your needs. Push and pull Docker images to your private Container Registry using the standard Docker command line interface. Use Docker naturally while searching images by name and tags.



## Fast, high-availability access

Using our regional private repositories across the world, you can get optimal response time globally. Store images close to your compute instances in Europe, Asia, or US and access Google's high-performing global network for fast deployment.

# GCP Container Registry



Google Cloud Platform test-kubernetes

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Container Registry

Repositories

test-kubernetes

Filter Enter property name or value

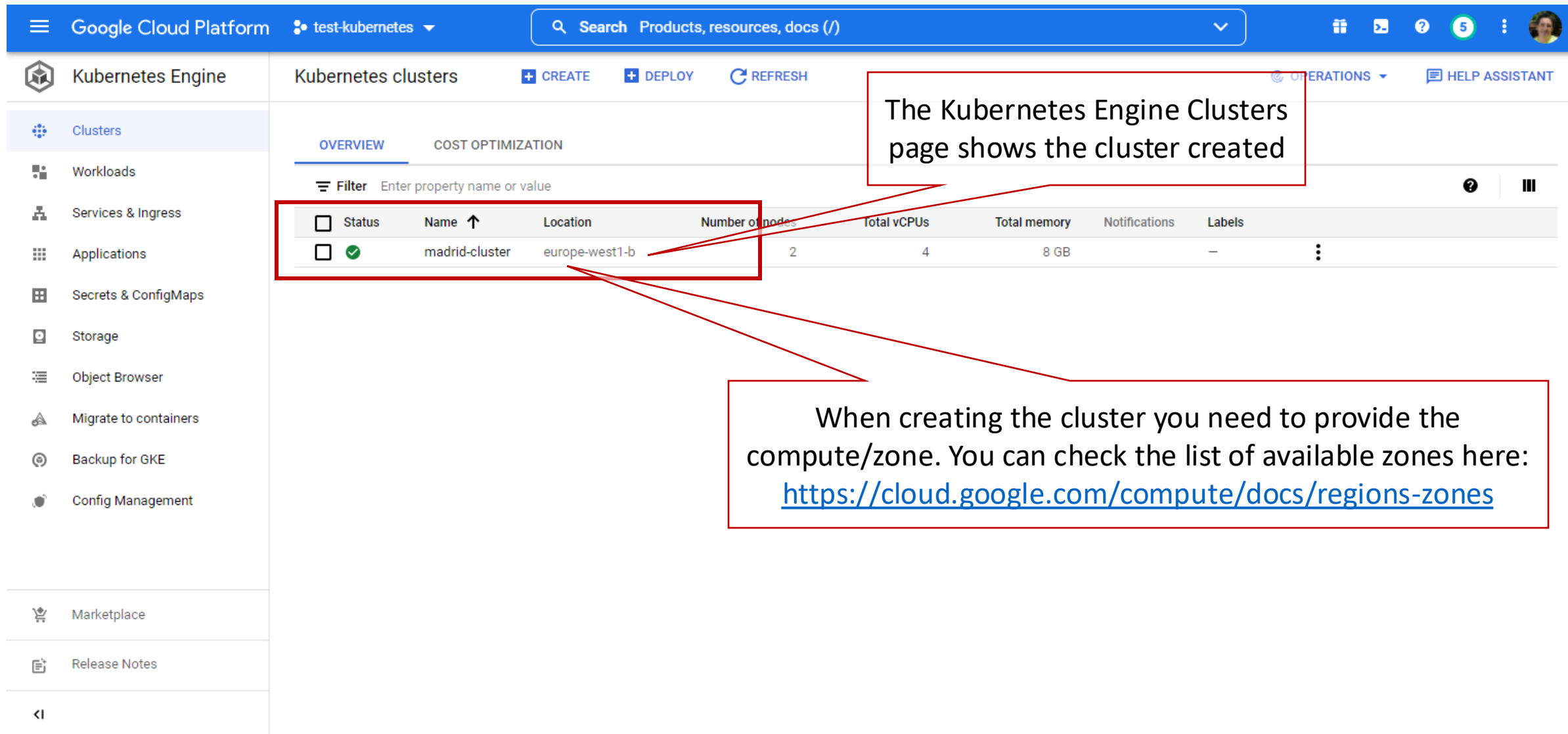
| Name ↑ | Hostname ? | Visibility ? |
|--------|------------|--------------|
| boston | gcr.io     | Private      |
| madrid | gcr.io     | Private      |

After pushing the image, In the Container Registry you can review your uploaded container images to gcr.io

# Google Kubernetes Engine

- What is Google Kubernetes Engine? <https://cloud.google.com/kubernetes-engine>
- Google Kubernetes Engine (GKE) provides a managed environment for deploying, managing, and scaling your containerized applications using Google infrastructure. The GKE environment consists of multiple machines (specifically, Compute Engine instances) grouped together to form a cluster.
- GKE clusters are powered by the **Kubernetes** open source cluster management system. Kubernetes provides the mechanisms through which you interact with your cluster. You use Kubernetes commands and resources to deploy and manage your applications, perform administration tasks, set policies, and monitor the health of your deployed workloads.

# GCP Kubernetes Engine Clusters



Google Cloud Platform test-kubernetes

Kubernetes Engine

Kubernetes clusters

CREATE DEPLOY REFRESH

OVERVIEW COST OPTIMIZATION

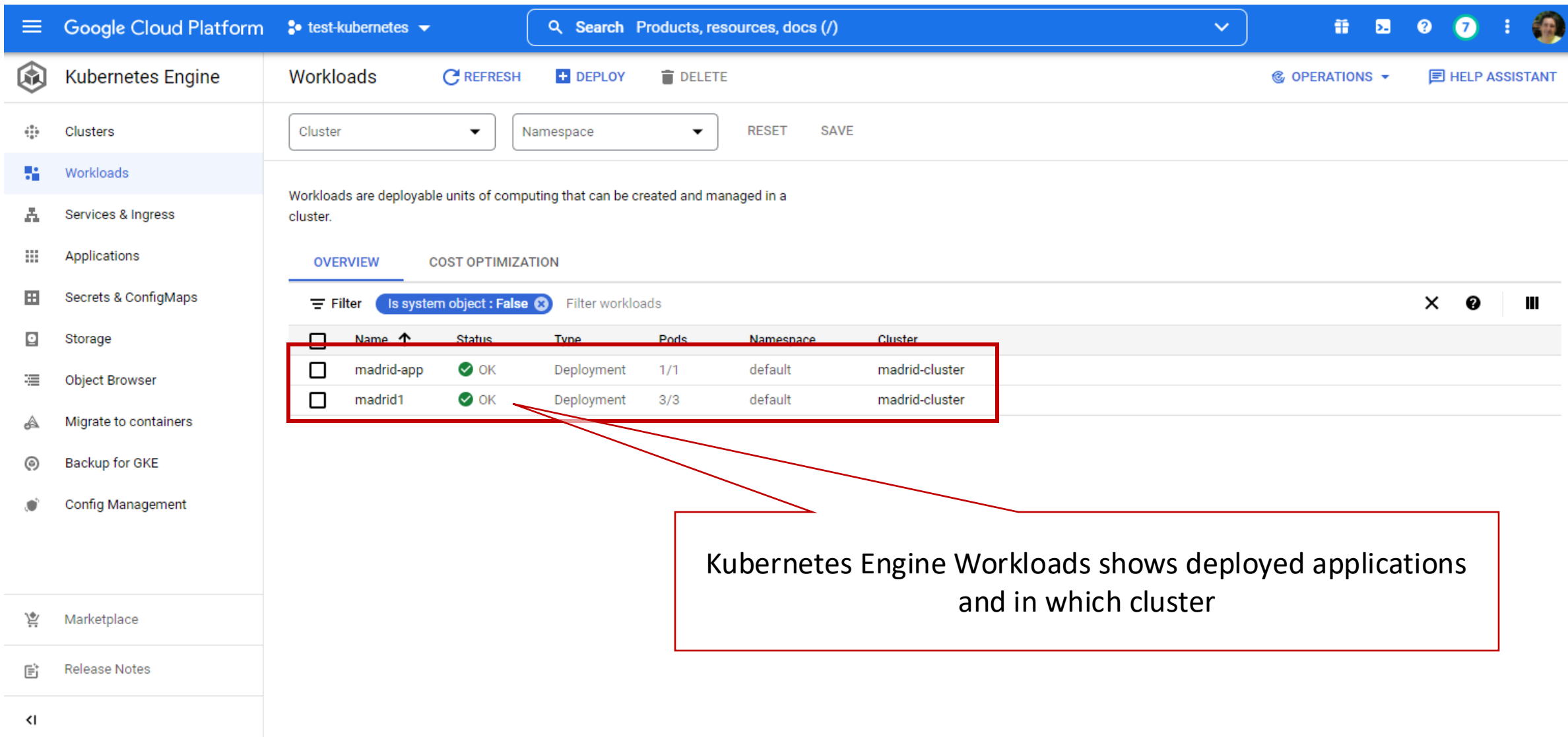
Filter Enter property name or value

| Status                              | Name ↑         | Location       | Number of nodes | Total vCPUs | Total memory | Notifications | Labels |
|-------------------------------------|----------------|----------------|-----------------|-------------|--------------|---------------|--------|
| <input checked="" type="checkbox"/> | madrid-cluster | europe-west1-b | 2               | 4           | 8 GB         | —             | ⋮      |

The Kubernetes Engine Clusters page shows the cluster created

When creating the cluster you need to provide the compute/zone. You can check the list of available zones here: <https://cloud.google.com/compute/docs/regions-zones>

# GCP Kubernetes Engine Workloads



Google Cloud Platform test-kubernetes

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Kubernetes Engine

Workloads REFRESH DEPLOY DELETE

OPERATIONS HELP ASSISTANT

Cluster Namespace RESET SAVE

Workloads are deployable units of computing that can be created and managed in a cluster.

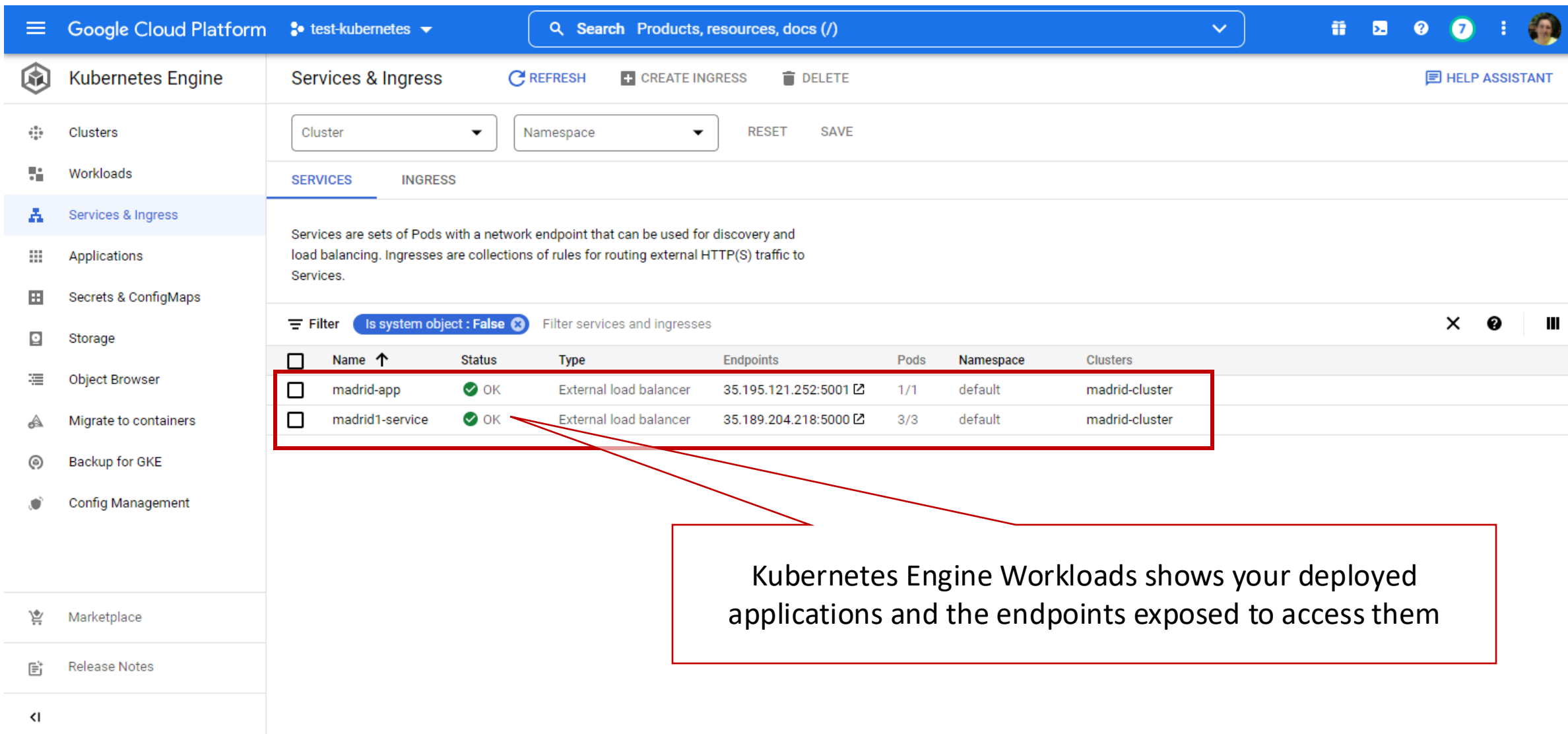
OVERVIEW COST OPTIMIZATION

Filter Is system object : False Filter workloads

| <input type="checkbox"/> | Name       | Status | Type       | Pods | Namespace | Cluster        |
|--------------------------|------------|--------|------------|------|-----------|----------------|
| <input type="checkbox"/> | madrid-app | OK     | Deployment | 1/1  | default   | madrid-cluster |
| <input type="checkbox"/> | madrid1    | OK     | Deployment | 3/3  | default   | madrid-cluster |

Kubernetes Engine Workloads shows deployed applications and in which cluster

# GCP Kubernetes Engine Services & Ingress



Google Cloud Platform test-kubernetes

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Kubernetes Engine Services & Ingress

Cluster Namespace RESET SAVE

SERVICES INGRESS

Services are sets of Pods with a network endpoint that can be used for discovery and load balancing. Ingresses are collections of rules for routing external HTTP(S) traffic to Services.

Filter Is system object : False Filter services and ingresses

| <input type="checkbox"/> | Name ↑          | Status | Type                   | Endpoints           | Pods | Namespace | Clusters       |
|--------------------------|-----------------|--------|------------------------|---------------------|------|-----------|----------------|
| <input type="checkbox"/> | madrid-app      | ✓ OK   | External load balancer | 35.195.121.252:5001 | 1/1  | default   | madrid-cluster |
| <input type="checkbox"/> | madrid1-service | ✓ OK   | External load balancer | 35.189.204.218:5000 | 3/3  | default   | madrid-cluster |

Kubernetes Engine Workloads shows your deployed applications and the endpoints exposed to access them

# Test model deployment

- Curl formatted for Windows CMD

```
c:\JM\Python_Projects\07_01 AWS Lambda>curl -X POST -H "Content-Type: application/json" -d "{\"crim\": 0.85204, \"zn\": 0, \"indus\": 8.14, \"chas\": 0, \"nox\": 0.538, \"rm\": 5.965, \"age\": 89.2, \"dis\": 4.0123, \"rad\": 4, \"tax\": 307, \"ptratio\": 21, \"lstat\": 13.83}" http://35.233.7.45:5000/predict  
{"Predicted House Price": 11.022853090683089}
```